

Intraday Liquidity Trading Playbook

Deep-research tutorial for opening drive continuation, liquidity sweeps, VWAP acceptance, strike/positioning reactions, and late-day continuation

Research synthesis of auctions, liquidity pockets, VWAP, options positioning, and intraday momentum

April 2026

How to read this manual

Educational use only. This guide is designed for liquid U.S. equities, ETFs, and index futures. It is a research synthesis and operational playbook—not personalized investment advice—and it assumes you will test, journal, and adapt the rules to your own execution quality, commissions, and risk tolerance.

Prepared as a standalone tutorial and playbook.

1. How to use this guide

This document does two jobs at once. First, it summarizes what the better source material actually says about opening and closing auctions, market depth, order-flow imbalance, VWAP, open interest, strike clustering, gamma effects, and intraday momentum. Second, it turns that research into a practical tutorial and trading playbook that you can apply by hand or adapt into a scanner, checklist, or backtest.

The core idea is simple: the best intraday trades usually happen where liquidity is concentrated, trapped, or suddenly withdrawn. In practice that means the open, prior highs and lows, premarket extremes, VWAP, obvious round numbers, and occasionally major option strikes or close-related imbalances.

Throughout the guide, the highest weight is given to exchange and regulatory documentation and to peer-reviewed academic work. Recent practitioner and SSRN papers are used only as suggestive evidence and are labeled that way.

Evidence quality note

Evidence ladder for this playbook: Tier A = exchange and regulatory documentation ([1]–[11]); Tier B = peer-reviewed academic work ([17]–[19]); Tier C = recent working papers and practitioner research ([20]–[22]).

Fast-start workflow

- Read Section 2 once to understand why these setups can exist.
- Use Section 3 as the map: pick the strategy that matches the day instead of memorizing every setup at once.
- Trade from Sections 4–8 only when the market condition matches the setup.
- Use Sections 9–11 as your operating system: regime filter, risk rules, and journaling.

Minimum viable tool stack

Chart / tool	Use
Daily chart	Context only: previous day levels, recent swing highs/lows, earnings gaps, major volume nodes.
5-minute chart	Primary structure chart for most stocks and ETFs.
1-minute chart	Execution chart for entries, pullbacks, and retests.
Session VWAP + volume	Core overlay for session acceptance plus participation.

Glossary

- ORB: opening range breakout.
- VWAP: volume-weighted average price.
- RVOL: relative volume versus a comparable time-of-day baseline.
- OI: open interest.
- R: a single unit of risk, defined by your stop distance and position size.

Intraday session map

Where each strategy tends to matter most in liquid U.S. equities, ETFs, and index futures



Suggested default to test: 5-minute opening range for stocks/ETFs and 1–5 minutes for index futures.

Figure 1. Session map showing when each setup tends to be most relevant.

2. Research foundation: why these setups can work

Opening and closing auctions are not background noise; they are formal price-discovery events. NYSE reported that its opening auction averaged 44 million shares and \$2.4 billion per day in the first four months of 2024, and its opening imbalance information provides a real-time view of paired quantity, imbalance quantity, and the book-clearing price [1]. NYSE imbalance messages are disseminated once per second during auctions [2]. Nasdaq likewise runs single-price opening and closing crosses and publishes imbalance information ahead of those auctions [3]. Nasdaq also states that its opening and closing auctions together account for more than 16% of daily trading volume on average, and more than 25% on ETF rebalance days [4].

The close matters even more than many discretionary traders appreciate. ICE says the NYSE Closing Auction remains the single largest daily liquidity event in U.S. cash equities trading, and in 2025 three of the largest NYSE closing auctions in history occurred, including a single-day auction record above \$205 billion in value traded [5]. Nasdaq describes the official close as the most important trade of the day because it anchors returns, valuations, settlements, tracking error, and index processes [6].

Liquidity pockets matter because price impact is not constant. Federal Reserve research explains that when market depth is low and liquidity is fragile, large directional order-flow imbalances have greater potential to amplify price moves [7]. The New York Fed notes that market depth is a common measure of liquidity because it captures how much size participants are willing to buy or sell at particular prices [8]. For traders, that means a breakout through a level is very different on a deep, orderly tape versus a thin tape where the book cannot absorb pressure.

VWAP is best understood first as an execution benchmark and only second as a chart overlay. SEC guidance on benchmark trades explicitly discusses VWAP as a benchmark price whose material terms may be determined over a specified period rather than off the current quote [9]. That is why the most useful way to trade around VWAP is not to treat it as a magical line, but as a proxy for whether the session is accepting above or below the average traded price so far.

Options and strike data are useful, but only as context. CME and IBKR define open interest as the number of contracts still open, carried over from the previous close, and emphasize that it can reveal where participation is concentrated by strike and expiry [12][13][14]. Academic research on expiration dates documents that closing prices of optionable stocks cluster around strike prices and

that returns are measurably altered on expiration dates [18]. But open interest alone does not tell you who is long or short, and Cboe's research stresses that high gross ODTE volume does not automatically imply large dealer hedging pressure. What matters is the net position and the sign of gamma: long gamma can dampen moves, while negative gamma can amplify them [15][16].

Intraday momentum is real enough to matter, especially when the day is already active. A Journal of Financial Economics paper found that the first half-hour return predicts the last half-hour return, with the effect stronger on volatile days, high-volume days, recession days, and major macro-news days [17]. That same logic helps explain why late-day continuation works best when the morning move was real, the midday tape did not destroy structure, and the close is approaching.

A final caution: backtests on opening-range and VWAP-based day-trading systems are not all created equal. Older published work on ORB strategies in crude oil futures found profitability in the full sample but weak robustness across subperiods [19]. More recent U.S. equity working papers report strong results when ORB is restricted to liquid "stocks in play" and when intraday momentum or VWAP systems are executed with discipline, but those papers remain working papers and their results are sensitive to costs, slippage, and selection rules [20][21][22]. Use them as inspiration, not as guarantees.

Operational takeaway

The playbook should be read as a sequence: map likely liquidity pockets, watch for acceptance or rejection, then choose the setup whose logic matches the tape. Do not reverse the process by deciding on a setup first and forcing the chart to fit it.

What not to assume

- VWAP is not predictive on its own; it is a session context tool.
- High open interest is not a guaranteed magnet or pin.
- High ODTE volume is not automatically destabilizing.
- The first move of the day is not mandatory to trade.
- A pass is a valid decision whenever execution quality is poor.

3. Strategy map at a glance

Strategy	Best regime	Primary trigger	Invalidation	Typical exit
Opening drive continuation	Catalyst + trend day	Opening-range break that holds; shallow pullback stays aligned with VWAP	Fast return into range or VWAP loss	Next liquidity pocket, then trail
Liquidity sweep & reclaim	Rotation / rejection day	Obvious level swept, then reclaimed back into prior range	Loss of reclaimed level or sweep extreme	VWAP, range midpoint, opposite side
VWAP acceptance / reclaim-failure	Trend resumption	First clean hold above/below VWAP after pullback	Flip back through VWAP with structure damage	Morning high/low, measured move
Strike / positioning reaction	Expiry / rebalance / major event	Repeated rejection or clean acceptance at	Failure to hold reaction level	Next strike, close, or VWAP

		major strike/positioning zone		
Late-day continuation	Strong morning trend survives lunch	Afternoon range break with VWAP still aligned	Back inside afternoon range	Into the close / closing auction

Market-specific adaptation

Product	What fits best	Notes
Index ETFs / futures	VWAP acceptance; late-day continuation; strike reaction	Best for clean execution and close-related flows.
Single-name catalyst stocks	Opening drive continuation; sweep & reclaim	Prioritize earnings, guidance, FDA, macro-sensitive leaders, or unusual news.
Slow, low-RVOL names	Usually pass	Most false signals come from forcing setups in poor participation.

Data tiers

Data access	Still usable?	What to rely on
Basic retail charts	Yes	Price, session VWAP, relative volume, premarket levels, prior day levels, major strikes / OI.
Advanced feeds	Better	Opening imbalance data, deeper order-book context, more reliable options positioning tools, close analytics.

Strategy selection decision tree

Route the session toward the right setup instead of forcing one strategy every day

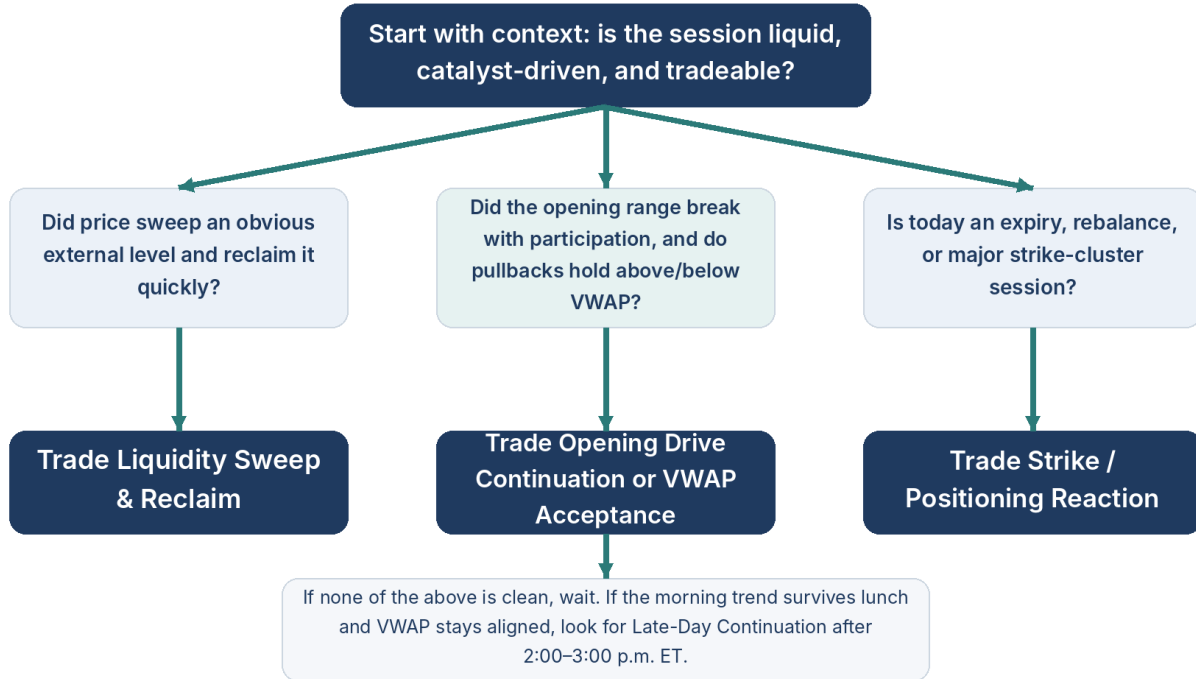


Figure 2. Strategy-selection decision tree.

3. Opening drive continuation

When the opening auction reveals real acceptance of overnight information and pullbacks stay orderly, the morning trend often extends. This setup is strongest in liquid names with a catalyst, broad participation, and shallow retracements. Auction documentation and intraday-momentum research both support giving the open unusual weight [1][2][3][4][17].

Setup card	
Best conditions	Liquid stock, ETF, or index future; catalyst or macro driver; clear relative-volume surge; aligned index/sector; shallow pullbacks; price stays outside the opening range.
Trigger	Break of the opening range followed by hold or break-retest-hold. Operational default: 5-minute range for stocks/ETFs, 1–5 minutes for index futures.
Initial stop	Inside the opening range or beyond the retest swing. If that stop is too wide for the spread and volatility, skip the trade.
Targets	Next obvious liquidity pocket: premarket high/low, prior day high/low, round number, or a measured move equal to the opening-range width.
Skip if	No catalyst, weak RVOL, repeated rejection back into the range, or spreads are too wide for clean entries.

Tutorial steps

1. Before the open, mark premarket high/low, previous day high/low/close, and the default opening-range window.
2. Let the first minutes print; do not predict direction on the first tick.
3. Require either a close outside the opening range or a break-retest-hold outside the range.
4. Enter on the first clean continuation after the retest, or on the first shallow pullback that remains aligned with VWAP.
5. Take partial profits into the next liquidity pocket. Hold a runner only if structure remains clean.

Confirmation checklist

- RVOL is clearly above normal for that time of day.
- VWAP slopes in the trade direction or at least is not instantly rejected.
- Pullbacks fail to retrace the full opening impulse.
- Sector/index behavior is not fighting the trade.

Common failure modes

- Chasing a move that is already extended before you can define risk.
- Treating every gap as a trend day even when the open immediately rotates back into value.
- Using market orders in names where the spread widens on every 1-minute bar.

Example scenario

Example scenario: a stock gaps up on earnings, prints a narrow 5-minute opening range above the prior day high, breaks out, retests the breakout shelf, and holds above VWAP. Entry is on the first higher low after the retest; the first exit is the next round number or premarket extension zone.

4. Liquidity sweep & reclaim

Obvious highs and lows attract resting stops and breakout orders. When price sweeps one of those pockets but cannot hold beyond it, trapped traders often fuel the move back into the prior range. This setup is a practical application of liquidity-pocket logic and order-flow imbalance: what matters is not the wick itself, but whether the market accepts the move after the sweep [7][8].

Setup card	
Best conditions	Rotation or rejection day; premarket or previous-day extreme nearby; extension into the level with poor follow-through; broad market not launching into a one-sided trend.
Trigger	Price takes the level, fails to hold, and reclaims the prior shelf or range. Stronger if VWAP also gets reclaimed (for longs) or lost again (for shorts).
Initial stop	Beyond the sweep extreme. If that distance is too large relative to the target back into range, pass.
Targets	VWAP, range midpoint, opposite side of the opening balance, or the next obvious external liquidity level.
Skip if	The market is in a confirmed trend day and the sweep is actually a real breakout with broad support.

Tutorial steps

6. Mark external liquidity first: premarket high/low, previous day high/low, obvious swing highs/lows, and key round numbers.
7. Wait for price to sweep the level. A wick alone is not enough.
8. Require a reclaim back inside the prior range or back through the reclaimed shelf.
9. Enter on the retest of the reclaimed level or on the first lower-risk continuation back into range.
10. Reduce size if the reclaim happens in a very thin tape with erratic spreads.

Confirmation checklist

- The reclaim happens quickly rather than hours later.
- The market cannot auction cleanly outside the swept level.
- The retest of the reclaimed level is defended.
- The setup is happening at a level other traders can realistically care about.

Common failure modes

- Shorting the first wick through a strong trend-day high without waiting for a reclaim.
- Trading a sweep in a low-participation midday tape where there is no follow-through either way.
- Ignoring the broader market when the whole tape is breaking out together.

Example scenario

Example scenario: SPY sweeps the premarket low during the first fifteen minutes, prints a sharp rejection, reclaims VWAP, and then holds the reclaimed intraday shelf. Entry is on the first higher low above VWAP; target one is range midpoint and target two is the opposite side of the opening balance.

5. VWAP acceptance / reclaim-failure

VWAP matters because institutions, execution algorithms, and post-trade evaluation all treat it as an important benchmark [9]. For discretionary traders, the edge comes not from the line itself but from what the market does around it: acceptance above VWAP suggests buyers control the session; rejection from below suggests sellers do.

Setup card	
Best conditions	After the first opening burst; price has already shown a directional bias; RVOL is still healthy; the tape is not crossing VWAP back and forth every few minutes.
Trigger	First clean hold above VWAP after a bullish pullback, or first clean failure below VWAP after a bearish bounce.
Initial stop	Beyond the local pivot on the far side of VWAP. If the stop must be too wide because price is whipping around the line, skip.
Targets	Morning high/low, measured move from the local balance, or a late-day hold if structure stays intact.
Skip if	VWAP is flat, price has already crossed it repeatedly, or the market has devolved into midday chop.

Tutorial steps

11. Use the open to determine whether there is a real directional bias first.

Educational use only — not investment advice.

12. Wait for the first organized pullback or bounce into VWAP.
13. Require either a hold-and-rotate higher (long) or a rejection-and-rotate lower (short).
14. Use the 1-minute chart for execution, but keep the 5-minute chart as the structure anchor.
15. If price starts crossing VWAP repeatedly, stop treating it as a tradable edge and move to pass mode.

Confirmation checklist

- Price respects the same side of VWAP on the retest.
- The local swing structure improves in the trade direction.
- Volume does not collapse completely on the retest.
- You can explain why this is acceptance or rejection, not just a random touch.

Common failure modes

- Buying every first touch of VWAP without context.
- Forcing VWAP setups when the session is structurally rotational.
- Ignoring nearby prior highs/lows or other liquidity pockets that matter more than VWAP in that moment.

Example scenario

Example scenario: QQQ trends higher in the morning, pulls back into a rising VWAP around 10:35 a.m., holds above it, and prints a higher low on the 1-minute chart. Entry is above the micro pullback high; first target is the morning high and the runner is trailed so long as price stays above VWAP.

6. Strike / positioning reaction trade

Heavy strike concentration and options positioning can matter, especially on expiration, rebalance, or major macro days—but only as context. Open interest shows where contracts are open [12][13][14]. Academic work documents strike clustering around expiration [18]. Cboe's research and recent ODTE work show that the direction and size of dealer hedging pressure depend on the sign and magnitude of net gamma, not on raw gross volume alone [15][16].

Setup card	
Best conditions	Index products, mega-cap names, expiry/rebalance days, macro event days, or sessions where price is repeatedly interacting with a major strike or positioning zone.
Trigger	Either repeated rejection from a heavy strike (pin/fade) or clean acceptance through it with volume and follow-through (through-and-go).
Initial stop	Based on local price structure around the strike—not the strike alone. Use the level only as context, not as the stop location by itself.
Targets	Next major strike, VWAP, the session high/low, or the close if the flow remains one-sided.
Skip if	You only have raw OI and no price confirmation; the tape is dead; or the name is too illiquid for options context to matter intraday.

Tutorial steps

16. Mark the major strikes with unusually high open interest or heavy same-day attention.

17. Ask whether the day's context makes those strikes more relevant: expiry, rebalance, macro, or strong index activity.
18. Watch how price behaves there: repeated rejection, chop and pinning, or decisive acceptance through the level.
19. Only enter once the tape confirms the path—fade the strike if price cannot accept away from it, or go with the move if price accepts through it.
20. Reassess as the close approaches, because strike and auction effects often matter more late than at noon.

Confirmation checklist

- The strike aligns with a visible chart level or round number that traders already care about.
- The reaction is visible in price and volume, not just in a positioning dashboard.
- You understand whether you are trading a pin/reversion or a through-and-go continuation.
- You are not assuming that OI alone reveals dealer direction.

Common failure modes

- Calling every heavy strike a magnet.
- Confusing gross option volume with net hedging pressure.
- Ignoring the possibility that positive gamma can dampen rather than amplify moves.

Example scenario

Example scenario: on a monthly expiration day, SPX spends much of the session oscillating around a large round strike. During the last hour it finally accepts above that level on expanding participation, holds the first retest, and begins targeting the next strike into the close. That is a through-and-go version of the setup, not a fade.

7. Late-day continuation

Late-day continuation is the cleanest way to apply intraday-momentum research. If the morning move was real, the midday period merely consolidated it, and VWAP stayed aligned, the final hour often resumes the trend. The setup is strongest because the close is a major benchmark and liquidity event [5][6][17].

Setup card	
Best conditions	Strong morning impulse; midday consolidation instead of reversal; VWAP still aligned; broad market and sector still confirm direction.
Trigger	Break of the afternoon range, usually after 2:00–3:00 p.m. ET, with price still on the correct side of VWAP.
Initial stop	Back inside the afternoon range or beyond the last consolidation pivot.
Targets	Into the close, the next external liquidity pocket, or the next major strike if options context matters that day.
Skip if	The day has already fully reversed the morning trend, VWAP has been lost repeatedly, or the afternoon range is directionless.

Tutorial steps

21. Label the morning impulse first. If there was no real morning impulse, you probably do not have a continuation candidate.
22. Watch the midday period: you want a sideways or shallow pullback, not a full structure break.
23. Mark the afternoon range high/low and wait for a real breakout attempt.
24. Require that VWAP still supports the trend and that the broader tape is not abruptly diverging.
25. Be more willing to carry the trade into the close than in other setups, because the close itself is part of the edge.

Confirmation checklist

- Morning trend remains visible on the 5-minute chart.
- Midday consolidation did not fully retrace the opening impulse.
- VWAP still agrees with the trend direction.
- The breakout happens before liquidity deteriorates into random end-of-day noise.

Common failure modes

- Treating a random afternoon pop in a range day as continuation.
- Ignoring large closing imbalances or broad-market reversals.
- Trying to force the setup after the session has already exhausted the day's true range.

Example scenario

Example scenario: ES trends up on a macro day, chops sideways from late morning through early afternoon without losing VWAP, and then breaks the afternoon range high after 2:30 p.m. ET. Entry is on the breakout or first tight retest; target is the close or the next external high.

8. Regime filter and daily operating procedure

Regime	What it looks like	Favor	De-emphasize
Trend day	One-sided morning impulse; shallow pullbacks; VWAP remains supportive or restrictive	Opening drive; VWAP acceptance; late-day continuation	Aggressive mean-reversion fades
Rotation day	Repeated flips around VWAP; failed breakouts; back-and-forth inside range	Liquidity sweep & reclaim; quicker targets	Blind breakout continuation
Fragile-liquidity day	Thin depth; air pockets; widening spreads; erratic follow-through	Smaller size or pass	Market orders and oversized stop distances
Macro / event day	Fast repricing around releases; sharp opening move; unusually high participation	Wait for structure, then focus on open and close	Predicting direction before the tape settles

Daily operating checklist

- Mark previous day high, low, and close.
- Mark premarket high and low, plus major round numbers.

Educational use only — not investment advice.

- Check catalyst, macro calendar, and whether the name is truly liquid.
- Note expected major strikes / OI clusters if options context is relevant.
- Choose the default opening-range window before the bell.
- After the open, classify the day: trend, rotation, fragile-liquidity, or macro event.
- Trade only the setups that fit the observed regime.

Playbook rule

Trade fewer setups, not more. Once the regime is clear, focus on the one or two setups that actually fit that regime and ignore the rest.

9. Risk management and execution rules

Most strategy failure comes from two places: trading the wrong regime and executing a valid idea with sloppy orders or oversized risk. The rules below are operational defaults—not laws of nature—but they are designed to keep the strategy edge from being destroyed by execution.

Rule	Why it matters
Define risk in R	Position size should be determined by stop distance, not by conviction. A useful operational default is to risk a fixed cash amount per trade and express results in R.
Use quality tiers	Example default: A-setup = 1.0R, B-setup = 0.5R, anything worse = pass. Context should change size, not stop discipline.
Respect execution reality	Prefer limit entries in volatile names. Understand that stop orders can become market orders and may fill far from the trigger price in fast conditions [10][11].
Set a daily loss cap	A practical default is to stop after 2 full-risk losses or after repeated process violations. The point is to prevent emotional regime-switching.
No averaging into invalidation	If the reason for the trade is wrong, size should go to zero—not higher.
Scale out into liquidity	Partial profits should usually be taken into the next obvious pocket instead of demanding a full trend day every time.

Execution caution

FINRA and Investor.gov both warn that stop orders can trigger on short-lived price moves and then execute at prices materially different from the stop in fast markets [10][11]. If you are trading volatile opens or thin names, assume your actual execution will be worse than your ideal chart price.

10. Journal template and setup scorecard

The fastest way to improve this playbook is to journal process, not just P&L. Every trade should leave behind a record of why it existed, why it was entered, and what invalidated it.

Field	Why it matters
-------	----------------

Setup name	Forces you to trade from the playbook rather than from impulse.
Catalyst / context	Tells you whether the trade had a reason to exist.
Regime classification	Shows whether the setup matched the session type.
Entry trigger	Lets you separate good ideas from poor execution.
Stop reason	Confirms that invalidation was structural, not arbitrary.
Exit reason	Shows whether you took profits into liquidity or gave back edge.
R result and process grade	Separates P&L luck from repeatable decision quality.

10-point setup scorecard

A simple operational default is to score each category from 0 to 2. Trade full size only when the total is strong enough to justify it.

Category	0	1	2
Catalyst / context	No reason	Mild reason	Clear catalyst / event
Participation	Poor RVOL	Acceptable RVOL	Strong RVOL / liquidity
Level clarity	Messy	Usable	Obvious / clean
Acceptance or rejection	Unclear	Mixed	Very clear
Execution quality	Wide / sloppy	Okay	Tight / controlled

Suggested sizing rule

8–10 points: A-setup, full size. 6–7 points: B-setup, half size. 0–5 points: pass. This is not a universal formula; it is a disciplined starting framework to test and refine.

11. One-page playbook summary

- Trade liquid products first. Execution quality is part of the edge.
- Map liquidity before the open: previous day levels, premarket extremes, VWAP, round numbers, and major strikes if relevant.
- Do not predict the first move. Let the open show acceptance or rejection.
- Breakouts need participation; sweeps need reclaims; VWAP needs structure.
- Use options positioning as context, not prophecy.
- The stronger the day, the more the first half-hour and last hour matter.
- Take partials into the next obvious liquidity pocket.
- When structure breaks, size should shrink to zero.
- The best setups are often obvious in hindsight because they were obvious in real time too.
- A pass is a position. Forced trades are not.

Bottom line

Map liquidity. Wait for acceptance or rejection. Trade only when context, participation, structure, and execution all agree. When they do not agree, the highest-quality decision is usually to stand aside.

References

The reference list below intentionally excludes any social-media or short-form video source and instead prioritizes exchange documentation, regulatory guidance, peer-reviewed research, and clearly labeled working papers.

Exchange, regulatory, and official market-structure sources

- [1] NYSE, "NYSE introduces the enhanced NYSE Auction tool with Opening Imbalance History" (2024).
- [2] NYSE, "Pillar Order Imbalances Feed Client Specification" (current client specification; auction imbalance messages disseminated once per second).
- [3] Nasdaq, "Opening and Closing Crosses" fact sheet / FAQ (single-price opening and closing cross mechanics and imbalance dissemination).
- [4] Nasdaq, "Data In Depth Webinar Series: Minding the Crosses" / Nasdaq Data page (auctions account for more than 16% of daily Nasdaq volume on average; ETF rebalance days can exceed 25%).
- [5] ICE / NYSE, "ICE Announces 2025 Records Across Its Global Derivative and NYSE Equity & NYSE Options Markets" (2026).
- [6] Nasdaq, "Market Close: The Most Important Price of the Day" (2026).
- [7] Federal Reserve Board, "Order Flow Imbalances and Amplification of Price Movements: Evidence from U.S. Treasury Markets" (2025).
- [8] Federal Reserve Bank of New York, Liberty Street Economics, "Measuring Treasury Market Depth" (2024).
- [9] SEC, "Responses to Frequently Asked Questions Concerning Rule 611 and Rule 610 of Regulation NMS" (VWAP benchmark FAQs).
- [10] FINRA, "Stop Orders: Factors to Consider During Volatile Markets" (2025).
- [11] Investor.gov, "Investor Bulletin: Understanding Order Types."
- [12] CME Group, "Open Interest" educational lesson.
- [13] Interactive Brokers, "Open Interest" lesson.
- [14] CME Group, "Open Interest Heatmap."
- [15] Cboe, Mandy Xu, "Volatility Insights: Much Ado About ODTEs — Evaluating the Market Impact of SPX ODTE Options" (2023).
- [16] García, Pearson, and Vasquez, "ODTE Index Options and Market Volatility: How Large is Their Impact?" (2025 working paper using Cboe trade data).

Peer-reviewed academic sources

- [17] Gao, Han, Li, and Zhou, "Market intraday momentum," *Journal of Financial Economics* (2018).
- [18] Ni, Pearson, and Poteshman, "Stock price clustering on option expiration dates," *Journal of Financial Economics* (2005).
- [19] Holmberg, Lönnbark, and Lundström, "Assessing the profitability of intraday opening range breakout strategies," *Review of Financial Economics* (2013).

Working papers and practitioner studies (suggestive, not definitive)

- [20] Zarattini, Barbon, and Aziz, "A Profitable Day Trading Strategy For The U.S. Equity Market" (SSRN working paper, 2024; revised 2025).
- [21] Zarattini, Aziz, and Barbon, "Beat the Market: An Effective Intraday Momentum Strategy for S&P500 ETF (SPY)" (SSRN working paper, 2024; revised 2025).
- [22] Zarattini and Aziz, "Volume Weighted Average Price (VWAP) — The Holy Grail for Day Trading Systems" (SSRN working paper, 2023; revised 2025).